10th Grade Midterm Analysis of Decisions: C4 Solutions

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Evaluated Criterion

C6: Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.

1 Introduction

Problems:

1.1. Expected Value Criterion

1.2. General 2x2 Problem

1.3. General 4x3 Problem

1.1 Expected Value Criterion

When probabilities can be assigned to the possible states of nature, a decision-maker can compare alternatives using their expected monetary values. The Expected Monetary Value (EMV) criterion, also known as the Bayes decision criterion, combines the payoff associated with each possible state of nature with the probability that the state will occur. For a payoff table with alternatives in rows and states of nature in columns, let $P_{ay}(A_i, S_j)$ represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

Let $P(S_j)$ represent the probability that state of nature S_j occurs.

The probabilities must satisfy

$$\sum_{j=1}^{n} P(S_j) = 1.$$

For each alternative, calculate its Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j) P_{ay}(A_i, S_j)$$

or, explicitly,

$$EMV(A_i) = P(S_1)P_{ay}(A_i, S_1) + P(S_2)P_{ay}(A_i, S_2) + \cdots + P(S_n)P_{ay}(A_i, S_n).$$

The Bayes decision is the alternative with the highest expected payoff:

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i).$$

Therefore, the decision-maker selects the alternative with the highest Expected Monetary Value.

1.2 General 2x2 Problem

Consider the following payoff table, where each state of nature has an associated probability:

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

The probabilities of the states of nature are

$$P(S_1), \qquad P(S_2), \qquad P(S_1) + P(S_2) = 1.$$

Determine the decision using the Expected Monetary Value criterion.

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2 States of nature: S_1, S_2 Probabilities $P(S_1), P(S_2)$

Step 2: Calculate the EMV of A_1

For alternative A_1, multiply each payoff by the probability of its corresponding state of nature and add the results:

$$EMV(A_1) = P(S_1)P_{ay}(A_1, S_1) + P(S_2)P_{ay}(A_1, S_2).$$

Thus, $EMV(A_1)$ represents the expected payoff from selecting alternative A_1.

Step 3: Calculate the EMV of A_2

For alternative A_2:

$$EMV(A_2) = P(S_1)P_{ay}(A_2, S_1) + P(S_2)P_{ay}(A_2, S_2).$$

Thus, $EMV(A_2)$ represents the expected payoff from selecting alternative A_2.

Step 4: Apply the Bayes decision criterion

Compare the two expected monetary values:

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i).$$

Therefore, choose the alternative with the highest Expected Monetary Value.

Extended payoff table

Alternative	S_1 ($P(S_1)$)	S_2 ($P(S_2)$)	EMV
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P(S_1)P_{ay}(A_1, S_1) + P(S_2)P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P(S_1)P_{ay}(A_2, S_1) + P(S_2)P_{ay}(A_2, S_2)$

1.3 General 4x3 Problem

Consider the following payoff table, where each state of nature has an associated probability:

Alternative	S_1	S_2	S_3
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$

The probabilities of the states of nature are

$$P(S_1), \qquad P(S_2), \qquad P(S_3), \qquad \text{with} \qquad P(S_1) + P(S_2) + P(S_3) = 1.$$

Determine the decision using the Expected Monetary Value criterion.

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2, A_3, A_4 States of nature: S_1, S_2, S_3

Probabilities: $P(S_1), P(S_2), P(S_3)$

Step 2: Calculate the EMV of A_1

$$EMV(A_1) = P(S_1)P_{ay}(A_1, S_1) + P(S_2)P_{ay}(A_1, S_2) + P(S_3)P_{ay}(A_1, S_3).$$

Step 3: Calculate the EMV of A_2

$$EMV(A_2) = P(S_1)P_{ay}(A_2, S_1) + P(S_2)P_{ay}(A_2, S_2) + P(S_3)P_{ay}(A_2, S_3).$$

Step 4: Calculate the EMV of A_3

$$EMV(A_3) = P(S_1)P_{ay}(A_3, S_1) + P(S_2)P_{ay}(A_3, S_2) + P(S_3)P_{ay}(A_3, S_3).$$

Step 5: Calculate the EMV of A_4

$$EMV(A_4) = P(S_1)P_{ay}(A_4, S_1) + P(S_2)P_{ay}(A_4, S_2) + P(S_3)P_{ay}(A_4, S_3).$$

Step 6: Apply the Bayes decision criterion

Compare the four expected monetary values:

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i).$$

Therefore, choose the alternative with the highest Expected Monetary Value.

Extended payoff table

Alternative	S_1 $P(S_1)$	S_2 $P(S_2)$	S_3 $P(S_3)$	EMV
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$	$P(S_1)P_{ay}(A_1, S_1) + P(S_2)P_{ay}(A_1, S_2) + P(S_3)P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$	$P(S_1)P_{ay}(A_2, S_1) + P(S_2)P_{ay}(A_2, S_2) + P(S_3)P_{ay}(A_2, S_3)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$	$P(S_1)P_{ay}(A_3, S_1) + P(S_2)P_{ay}(A_3, S_2) + P(S_3)P_{ay}(A_3, S_3)$
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$	$P(S_1)P_{ay}(A_4, S_1) + P(S_2)P_{ay}(A_4, S_2) + P(S_3)P_{ay}(A_4, S_3)$

2 Contextualized Problems

Problems:

2.1. 2x2 Problem - Coffee shop

2.2. 2x2 Problem - Clothing store

2.3. 4x3 Problem - Technology company

2.4. 3x4 Problem - Logistics company

2.1 2x2 Problem - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager estimates that there is a 60% probability of high customer traffic during the promotional weekend and a 40% probability of low customer traffic. All payoffs are expressed in thousands of dollars. What would a perfectly logical decision-maker choose using the Expected Monetary Value criterion?

Alternative	High traffic S_1 $P(S_1) = 0.60$	Low traffic S_2 $P(S_2) = 0.40$
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch dessert line, A_2 = Keep current menu

States of nature: S_1 = High traffic, S_2 = Low traffic

Probabilities: $P(S_1) = 0.60, P(S_2) = 0.40$

Step 2: Calculate the EMV of A_1

For the Launch dessert line alternative:

$$EMV(A_1) = P(S_1)(42) + P(S_2)(8)$$

$$= (0.60)(42) + (0.40)(8)$$

$$= 25.2 + 3.2 = 28.4.$$

Thus, the expected payoff from launching the dessert line is 28.4 thousand dollars.

Step 3: Calculate the EMV of A_2

For the Keep current menu alternative:

$$EMV(A_2) = P(S_1)(28) + P(S_2)(20)$$

$$= (0.60)(28) + (0.40)(20)$$

$$= 16.8 + 8 = 24.8.$$

Thus, the expected payoff from keeping the current menu is 24.8 thousand dollars.

Step 4: Apply the Bayes decision criterion

Compare the two expected monetary values:

$$EMV(A_1) = 28.4, \qquad EMV(A_2) = 24.8.$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = A_1.$$

Thus, the Bayes criterion recommends launching the dessert line because it has the highest Expected Monetary Value, with an expected payoff of 28.4 thousand dollars.

Extended payoff table Extended payoff table

Alternative	High traffic S_1 $P(S_1) = 0.60$	Low traffic S_2 $P(S_2) = 0.40$	EMV
A_1: Launch dessert line	42	8	$0.60(42) + 0.40(8) = 28.4$
A_2: Keep current menu	28	20	$0.60(28) + 0.40(20) = 24.8$

2.2 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager estimates that there is a 70% probability of strong sales during the holiday season and a 30% probability of weak sales. All payoffs are expressed in thousands of dollars.

What would a perfectly logical decision-maker choose using the Expected Monetary Value criterion?

Alternative	Strong sales S_1 $P(S_1) = 0.70$	Weak sales S_2 $P(S_2) = 0.30$
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch new campaign, A_2 = Keep current strategy

States of nature: S_1 = Strong sales, S_2 = Weak sales

Probabilities: $P(S_1) = 0.70, P(S_2) = 0.30$

Step 2: Calculate the EMV of A_1

For the Launch new campaign alternative:

$$EMV(A_1) = P(S_1)(55) + P(S_2)(5)$$

$$= (0.70)(55) + (0.30)(5)$$

$$= 38.5 + 1.5 = 40.$$

Thus, the expected payoff from launching the new campaign is 40 thousand dollars.

Step 3: Calculate the EMV of A_2

For the Keep current strategy alternative:

$$EMV(A_2) = P(S_1)(35) + P(S_2)(22)$$

$$= (0.70)(35) + (0.30)(22)$$

$$= 24.5 + 6.6 = 31.1.$$

Thus, the expected payoff from keeping the current strategy is 31.1 thousand dollars.

Step 4: Apply the Bayes decision criterion

Compare the two expected monetary values:

$$EMV(A_1) = 40, \qquad EMV(A_2) = 31.1.$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = A_1.$$

Thus, the Bayes criterion recommends launching the new campaign because it has the highest Expected Monetary Value, with an expected payoff of 40 thousand dollars.

Extended payoff table

Alternative	S_1 $P(S_1) = 0.70$	S_2 $P(S_2) = 0.30$	EMV
A_1: Launch new campaign	55	5	$0.70(55) + 0.30(5) = 40$
A_2: Keep current strategy	35	22	$0.70(35) + 0.30(22) = 31.1$

2.3 4x3 Problem - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager estimates that there is a 50% probability of high demand, a 30% probability of medium demand, and a 20% probability of low demand. The company is considering four possible strategies: launching a premium version, launching a standard version, launching a budget version, or delaying the launch. The estimated profit for each strategy under each possible market condition is shown in the table below. All payoffs are expressed in thousands of dollars.

What would a perfectly logical decision-maker choose using the Expected Monetary Value criterion?

Alternative	High demand S_1 $P(S_1) = 0.50$	Medium demand S_2 $P(S_2) = 0.30$	Low demand S_3 $P(S_3) = 0.20$
Premium version (A_1)	80	35	-20
Standard version (A_2)	60	42	18
Budget version (A_3)	40	32	25
Delay launch (A_4)	15	15	15

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Premium version, A_2 = Standard version, A_3 = Budget version, A_4 = Delay launch

States of nature: S_1 = High demand, S_2 = Medium demand, S_3 = Low demand

Probabilities: $P(S_1) = 0.50, P(S_2) = 0.30, P(S_3) = 0.20$

Step 2: Calculate the EMV of A_1 and A_2

For the Premium version alternative:

$$EMV(A_1) = P(S_1)(80) + P(S_2)(35) + P(S_3)(-20)$$

$$= (0.50)(80) + (0.30)(35) + (0.20)(-20)$$

$$= 40 + 10.5 - 4 = 46.5$$

For the Standard version alternative:

$$EMV(A_2) = P(S_1)(60) + P(S_2)(42) + P(S_3)(18)$$

$$= (0.50)(60) + (0.30)(42) + (0.20)(18)$$

$$= 30 + 12.6 + 3.6 = 46.2$$

Step 3: Calculate the EMV of A_3 and A_4

For the Budget version alternative:

$$EMV(A_3) = P(S_1)(40) + P(S_2)(32) + P(S_3)(25)$$

$$= (0.50)(40) + (0.30)(32) + (0.20)(25)$$

$$= 20 + 9.6 + 5 = 34.6$$

For the Delay launch alternative:

$$EMV(A_4) = P(S_1)(15) + P(S_2)(15) + P(S_3)(15)$$

$$= (0.50)(15) + (0.30)(15) + (0.20)(15)$$

$$= 7.5 + 4.5 + 3 = 15$$

Step 4: Apply the Bayes decision criterion

Compare the four expected monetary values:

$$EMV(A_1) = 46.5 \qquad EMV(A_2) = 46.2 \qquad EMV(A_3) = 34.6 \qquad EMV(A_4) = 15$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = A_1$$

Thus, the Bayes criterion recommends launching the premium version because it has the highest Expected Monetary Value, with an expected payoff of 46.5 thousand dollars.

Extended payoff table

Alternative	S_1 $P(S_1) = 0.5$	S_2 $P(S_2) = 0.3$	S_3 $P(S_3) = 0.2$	EMV
A_1: Premium version	80	35	-20	$0.5(80) + 0.3(35) + 0.2(-20) = 46.5$
A_2: Standard version	60	42	18	$0.5(60) + 0.3(42) + 0.2(18) = 46.2$
A_3: Budget version	40	32	25	$0.5(40) + 0.3(32) + 0.2(25) = 34.6$
A_4: Delay launch	15	15	15	$0.5(15) + 0.3(15) + 0.2(15) = 15$

2.4 3x4 Problem - Logistics company

A logistics company is planning its warehouse capacity for the next year. The manager expects demand for storage services to vary depending on the level of activity in the market. Based on current forecasts, there is a 20% probability of very low demand, a 30% probability of low demand, a 35% probability of high demand, and a 15% probability of very high demand.

The company is considering three possible capacity strategies: renting a small warehouse, renting a medium warehouse, or renting a large warehouse. The estimated annual profit for each strategy under each possible demand level is shown in the table below. All payoffs are expressed in thousands of dollars.

What would a perfectly logical decision-maker choose using the Expected Monetary Value criterion?

Alternative	Very low demand S_1 $P(S_1) = 0.20$	Low demand S_2 $P(S_2) = 0.30$	High demand S_3 $P(S_3) = 0.35$	Very high demand S_4 $P(S_4) = 0.15$
Small warehouse (A_1)	25	35	48	55
Medium warehouse (A_2)	5	32	65	82
Large warehouse (A_3)	-25	15	72	110

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Small warehouse, A_2 = Medium warehouse, A_3 = Large warehouse

States of nature: S_1 = Very low demand, S_2 = Low demand, S_3 = High demand, S_4 = Very high demand

Probabilities: $P(S_1) = 0.20, P(S_2) = 0.30, P(S_3) = 0.35, P(S_4) = 0.15$

Step 2: Calculate the EMV of A_1 and A_2

For the Small warehouse alternative:

$$EMV(A_1) = P(S_1)(25)+P(S_2)(35)+P(S_3)(48)+P(S_4)(55)$$

$$= (0.20)(25) + (0.30)(35) + (0.35)(48) + (0.15)(55)$$
$$= 5 + 10.5 + 16.8 + 8.25 = 40.55$$

For the Medium warehouse alternative:

$$EMV(A_2) = P(S_1)(5)+P(S_2)(32)+P(S_3)(65)+P(S_4)(82)$$
$$= (0.20)(5) + (0.30)(32) + (0.35)(65) + (0.15)(82)$$
$$= 1 + 9.6 + 22.75 + 12.3 = 45.65$$

Step 3: Calculate the EMV of A_3

For the Large warehouse alternative:

$$EMV(A_3) = P(S_1)(-25)+P(S_2)(15)+P(S_3)(72)+P(S_4)(110)$$
$$= (0.20)(-25) + (0.30)(15) + (0.35)(72) + (0.15)(110)$$
$$= -5 + 4.5 + 25.2 + 16.5 = 41.2$$

Step 4: Apply the Bayes decision criterion

Compare the three expected monetary values:

$$EMV(A_1) = 40.55 \qquad EMV(A_2) = 45.65 \qquad EMV(A_3) = 41.2$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = A_2$$

Thus, the Bayes criterion recommends renting the medium warehouse because it has the highest Expected Monetary Value, with an expected annual payoff of 45.65 thousand dollars.

Extended payoff table

State	A_1: Small warehouse	A_2: Medium warehouse	A_3: Large warehouse	Probability
S_1	25	5	-25	0.2
S_2	35	32	15	0.3
S_3	48	65	72	0.35
S_4	55	82	110	0.15
EMV	$0.2(25)$ $+0.3(35)$ $+0.35(48)$ $+0.15(55)$ $= 40.55$	$0.2(5)$ $+0.3(32)$ $+0.35(65)$ $+0.15(82)$ $= 45.65$	$0.2(-25)$ $+0.3(15)$ $+0.35(72)$ $+0.15(110)$ $= 41.2$	0.2 $+0.3$ $+0.35$ $+0.15$ $= 1$